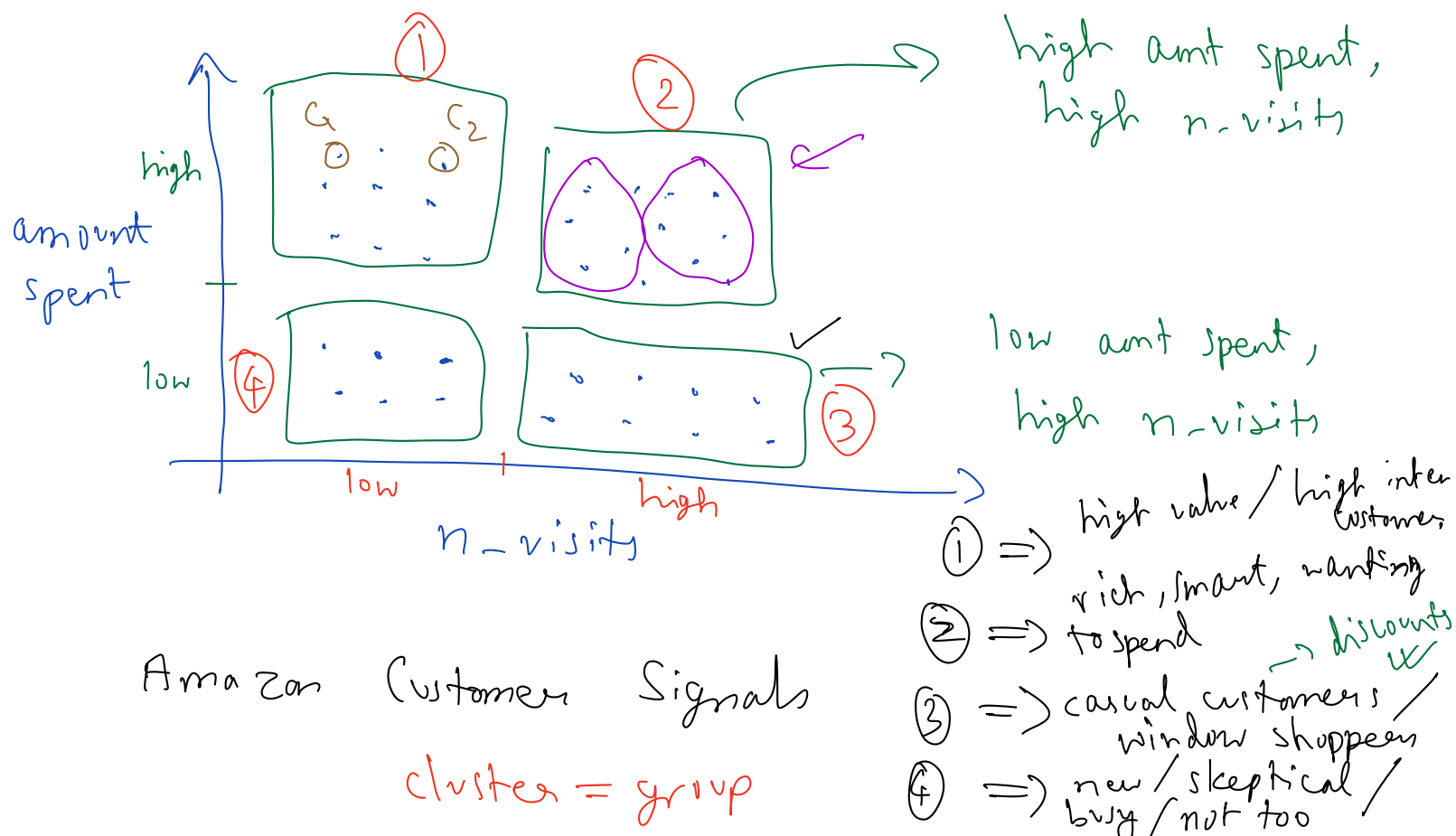


Today's agenda

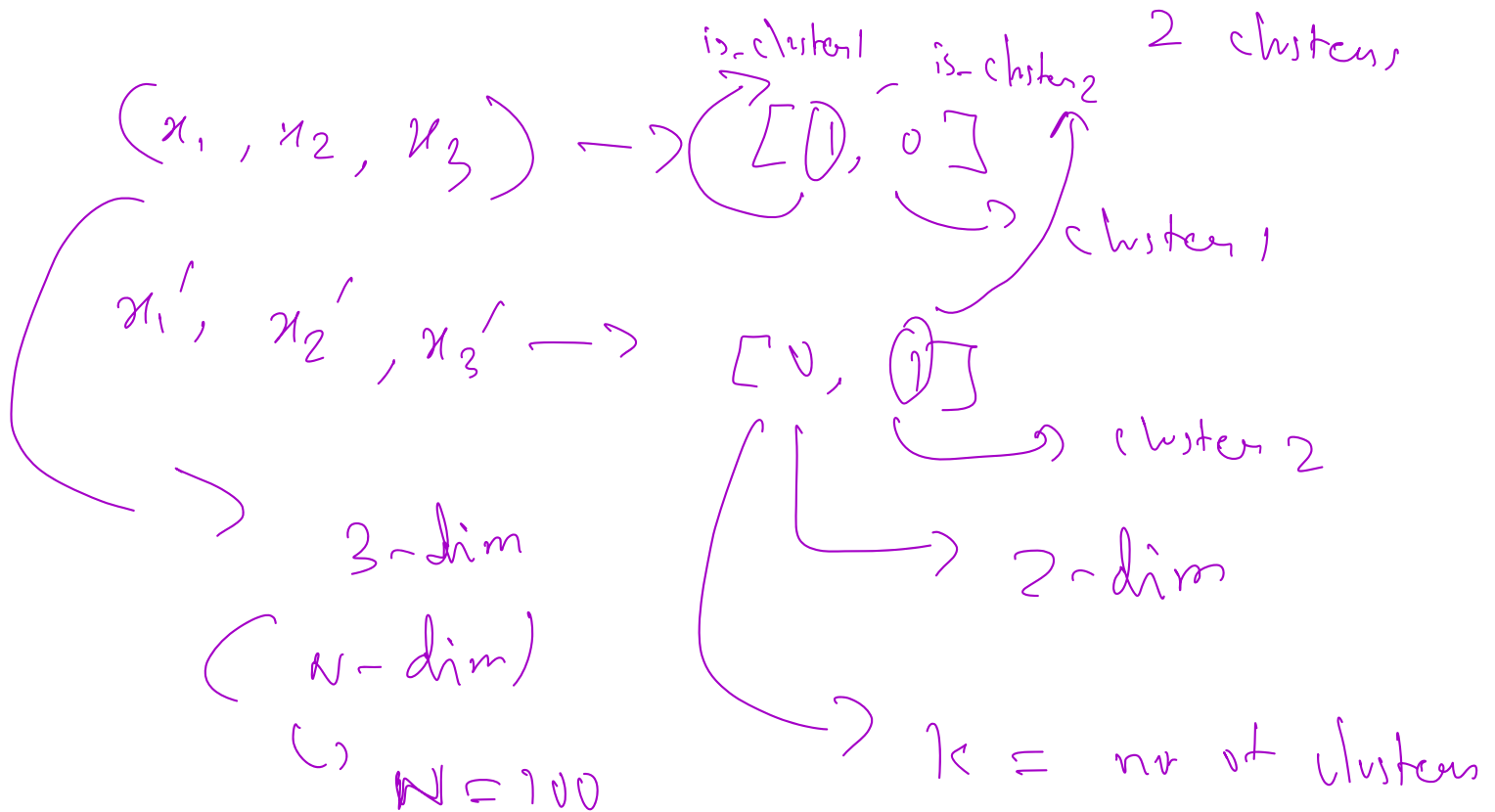
- 1) Intro to Unsupervised Learning & Problem Statement
- 2) Clustering & intuition
- 3) Dunn Index
- 4) k-Means & its Math Formulation
- 5) Lloyd's Algorithm
- 6) k-Means Scratch Implementation
- 7) Determining k
- 8) Amazon's customer segmentation with k-Means



$$2 \times 2 \Rightarrow 3 \times 3$$

interested
↳ discounts ??

low, med, high



① ② ③ ④ ⑤ ⑥ ⑦
 $[0.5, -1.7, 2.1, 100.5, 7.9, -1000, 84]$

7-dim

1,000,000 data-points with 7-dims each

4 clusters

cluster

1

2

3

4

$\rightarrow [1, 0, 0, 0]$

$\rightarrow [0, 0, 1, 0]$

$\rightarrow [0, 0, 0, 1]$

1-dim

$$I/P: \quad x_1: [x_{11}, x_{12}, x_{13}, \dots, x_{1d}] \quad \hookrightarrow d\text{-dim}$$

x_2

x_3

$\vdots$

$$x_N: [x_{N1}, x_{N2}, x_{N3}, \dots, x_{Nd}]$$

N as large as $1M$

$\hookrightarrow d\text{-dim}$

$$O/P: \quad x_1 \rightarrow \text{cluster no. } (1/2/3/\dots/k)$$

$\vdots$

$$x_N \rightarrow \text{cluster no. } (1/2/3/\dots/k)$$

$$c_1, c_2, \dots, c_k \rightarrow d\text{-dim}$$

Lloyd's algo

1) Randomisation: Randomly select k - points from N total points & assign them as k cluster centroids

$$c_1 \rightarrow x_{100}$$

$$c_2 \rightarrow x_1$$

$\vdots$

$$c_k \rightarrow x_{100}$$



1,000,000

data - points

2) Based on $[c_1, c_2, \dots, c_k]$, x_i will be assigned to cluster no.

$x_i \rightarrow \begin{matrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{matrix} \Bigg] \text{ min distance between } x_i \text{ \& } c_j$

Assignment

$\hookrightarrow j$ is your cluster no.
 $x_i = j$ (cluster value)

3) Re-compute the centroid

cluster 1: $x_{100}, x_2, x_{1000}, x_{5000}, \dots$

$\hookrightarrow c_1 = \text{avg}(x_{100}, x_2, x_{1000}, x_{5000}, \dots)$

$c_2 = \text{avg}(\dots)$

$\vdots$

$c_k =$

(Centroid computation)

$c_1(100) \longleftrightarrow c_1(101)$
 $\hookrightarrow \text{dist} \rightarrow \text{Very low}$

$C_3 (100)$ $\longleftrightarrow$ $C_3 (100)$
 $\searrow$ distance is very low

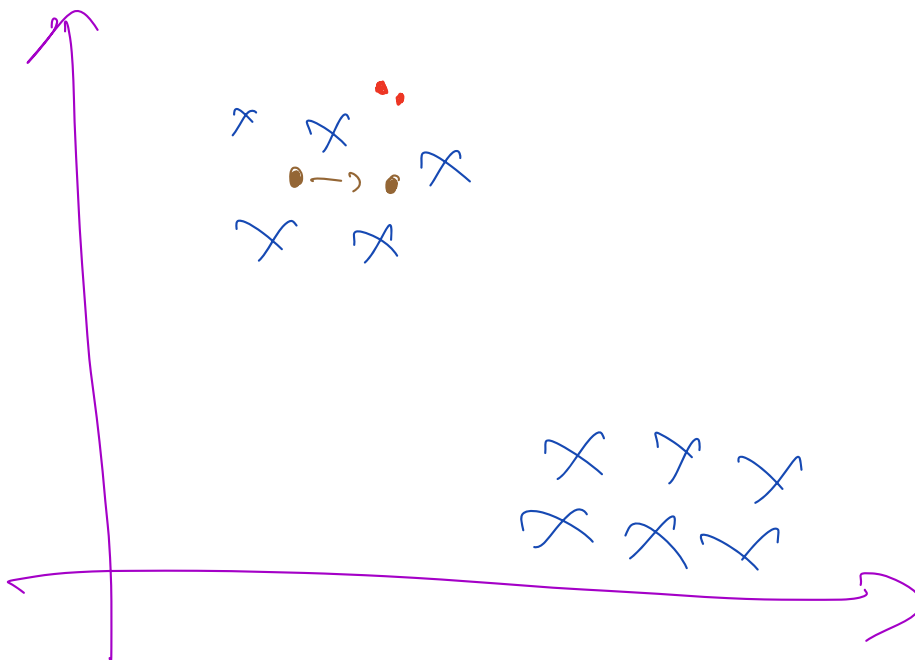
iteration 1 :

x_1, x_2
 $\searrow$ $C_1 (1)$

iteration 2 :

$x_3, x_1 \rightarrow C_1 (2)$

$\nearrow$ different



WCSS

(within cluster sum of squares)

$$= \sum_{i=1}^{m_1} (x_i - c_1)^2 + \sum_{i=1}^{m_2} (x_i - c_2)^2 + \dots$$

all points in cluster 1

all points in cluster 2

min WCSS

